

A sharp complex Dvoretzky theorem under phase-increment regularity

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23 July 2026

Status. Substantial partial result; current verdict: likely valid. The unrestricted removal of circledness remains open.

1 Source question

Write $S_{\mathbb{C}^n}$ for the Euclidean unit sphere in $\mathbb{C}^n$. Aubrun, Szarek, and Werner [1, Theorem 4] prove that if $f : S_{\mathbb{C}^n} \rightarrow \mathbb{R}$ is 1-Lipschitz and *circled*,

$$f(\omega z) = f(z) \quad (\omega \in \mathbb{T}),$$

then a random complex subspace E of dimension $cn\varepsilon^2$ satisfies

$$\sup_{z \in S_E} |f(z) - \mu| \leq \varepsilon$$

with high probability, where μ is a central value. They ask whether the circledness assumption can be removed completely. The exact source passage, on page 6, is reproduced below.

the fact that an ε -net in $S_{\mathbb{C}^n} = S^{2n-1}$ can be chosen to have cardinality $\leq (1 + 2/\varepsilon)^{2n}$ (see [14], Lemma 4.10). Indeed, this was essentially Milman’s original argument in [10]. However, one only obtains this way a subspace E of dimension $cn\varepsilon^2/\log(1/\varepsilon)$. For many applications (including our previous paper [9]), this extra logarithmic factor is not an issue. However, in the present case, having the optimal dependence on ε is crucial.

The classical framework of convex geometry is the real case (with or without the assumption “circled,” which in that context just means then that the function is even). In that setting, Theorem 4 was proved by Gordon [15] who used comparison inequalities for Gaussian processes. A proof based on concentration of measure was later given by Schechtman [16]. The complex case does not seem to appear in the literature. Actually, at the face of it, Gordon’s proof does not extend to the complex setting, while Schechtman’s proof does. We sketch Schechtman’s proof of Theorem 4 in Appendix A. It is not clear whether the assumption “ f circled” in Theorem 4 can be completely removed; we do know that it is needed at most for very small values of ε .

PROOF OF THE MAIN PROPOSITION

Let S_{HS} be the Hilbert–Schmidt sphere in $\mathcal{M}_{k,d}$ and let M be a random matrix uniformly distributed on S_{HS} . Let $\tilde{g}(\cdot)$ be the function defined on S_{HS} by

$$\tilde{g}(M) = \left\| MM^\dagger - \frac{\text{Id}}{k} \right\|_{HS}.$$

The later monograph of Aubrun and Szarek [2, Exercise 7.8] removes circledness above a logarithmic small- ε threshold and still records the removal of that threshold as unknown. The theorem below instead identifies a quantitative phase-regularity condition under which the sharp $n\varepsilon^2$ dimension holds at every scale allowed by the dimension.

2 Proof intuition

Schechtman's increment argument works whenever the two input vectors have nonnegative real inner product. Circledness is used only to rotate one vector into that position without changing the function value. For a non-circled function, retain the rotation but keep the resulting error as a second increment. The first increment is controlled by the source's conditional Lévy argument; the second is a phase increment on one Hopf circle.

If phase increments have Lipschitz constant proportional to the phase chord, Lévy concentration controls the second term at exactly the missing scale. The whole process is then subgaussian for the ordinary Euclidean metric, so the original Dudley chaining proof goes through. A $C^{1,1}$ bound supplies the required phase regularity by a direct gradient calculation.

3 Phase-increment regularity

All Lipschitz constants below use the Euclidean chord metric. For an L -Lipschitz function $f : S_{\mathbb{C}^n} \rightarrow \mathbb{R}$, define

$$D_{\mathbb{T}}(f) = \sup_{\omega \in \mathbb{T} \setminus \{1\}} \frac{\text{Lip}(z \mapsto f(\omega z) - f(z))}{|\omega - 1|} \in [0, \infty]. \quad (1)$$

Circled functions have $D_{\mathbb{T}}(f) = 0$, but finite phase modulus is much less restrictive than circledness.

Theorem 1. *There are absolute constants $c, C > 0$ with the following property. Let $n \geq 2$, let $f : S_{\mathbb{C}^n} \rightarrow \mathbb{R}$ be L -Lipschitz, suppose $D = D_{\mathbb{T}}(f) < \infty$, and set $K = L + D$. Let*

$$\mu = \int_{S_{\mathbb{C}^n}} f \, d\sigma.$$

If $E \subset \mathbb{C}^n$ is a Haar-distributed k -dimensional complex subspace and

$$1 \leq k \leq cn \frac{\varepsilon^2}{K^2}, \quad (2)$$

then

$$\mathbb{P} \left\{ \sup_{z \in S_E} |f(z) - \mu| > \varepsilon \right\} \leq C \exp \left(-cn \frac{\varepsilon^2}{K^2} \right).$$

If $K = 0$, then f is constant and the assertion is immediate; below we implicitly consider $K > 0$.

Remark 2. Changing the mean to a median or any central value only changes absolute constants. Indeed, Lévy concentration implies that any two such values differ by $O(L/\sqrt{n})$, while (2) with $k \geq 1$ forces $\varepsilon \geq c^{-1/2}K/\sqrt{n}$.

4 Subgaussian increments

We use the standard consequence of Lévy's lemma that if $h : S_{\mathbb{C}^q} \rightarrow \mathbb{R}$ has mean zero and Lipschitz constant A , then

$$\mathbb{P}\{|h| > t\} \leq C \exp(-cq t^2/A^2). \quad (3)$$

Lemma 3 (Aligned increment). *Let $x, y \in S_{\mathbb{C}^n}$ satisfy $\langle x, y \rangle \in [0, 1]$, and let U be Haar-distributed on $U(n)$. Then*

$$\mathbb{P}\{|f(Ux) - f(Uy)| > t\} \leq C \exp\left(-cn \frac{t^2}{L^2|x - y|^2}\right).$$

Proof. If $x = y$, the assertion is immediate. Thus $\beta > 0$ below. Put $z = (x + y)/2$, $w = (x - y)/2$, and $\beta = |w| = |x - y|/2$. The hypothesis makes $z \perp w$. Conditionally on $u = Uz$, the vector $U(w/\beta)$ is uniform on the unit sphere of the complex orthogonal complement of u . On that sphere consider

$$h_u(v) = f(u + \beta v) - f(u - \beta v).$$

The substitution $v \mapsto -v$ shows that h_u has conditional mean zero, and its Lipschitz constant is at most $2L\beta = L|x - y|$. Apply (3) conditionally and then integrate in u . This is the increment argument used in the source proof. $\square$

Lemma 4 (Phase alignment). *For every $x, y \in S_{\mathbb{C}^n}$, one can choose $\omega \in \mathbb{T}$ such that*

$$\langle x, \omega y \rangle = |\langle x, y \rangle|, \quad |x - \omega y| \leq |x - y|, \quad |\omega - 1| \leq \sqrt{2}|x - y|.$$

Proof. If $\langle x, y \rangle = 0$, take $\omega = 1$. Otherwise choose the phase which makes the inner product positive. Writing $\langle x, y \rangle = re^{i\theta}$, the first two assertions follow from

$$|x - y|^2 = 2 - 2r \cos \theta, \quad |x - \omega y|^2 = 2 - 2r.$$

Also $|\omega - 1|^2 = 2 - 2 \cos \theta$. If $\cos \theta \geq 0$, this is at most $|x - y|^2$; if $\cos \theta < 0$, it is at most 4, while $|x - y|^2 \geq 2$. $\square$

Lemma 5 (General increment). *For $x, y \in S_{\mathbb{C}^n}$,*

$$\mathbb{P}\{|f(Ux) - f(Uy)| > t\} \leq C \exp\left(-cn \frac{t^2}{K^2|x - y|^2}\right).$$

Proof. Choose ω from Lemma 4 and split

$$\begin{aligned} f(Ux) - f(Uy) &= [f(Ux) - f(U\omega y)] \\ &\quad + [f(U\omega y) - f(Uy)]. \end{aligned}$$

Lemma 3 controls the first term at subgaussian scale $L|x - \omega y|/\sqrt{n}$. For the second, the function

$$g_\omega(z) = f(\omega z) - f(z)$$

has mean zero by rotational invariance of σ , and by definition its Lipschitz constant is at most $D|\omega - 1|$. Equation (3) controls it at scale $D|\omega - 1|/\sqrt{n}$. The two chord estimates in Lemma 4, followed by a union bound, give the result after changing absolute constants. $\square$

5 Chaining and proof of the theorem

Proof of Theorem 1. Write $E = U(\mathbb{C}^k)$ and consider the centered process

$$X_z = f(Uz) - \mu, \quad z \in S_{\mathbb{C}^k}.$$

It satisfies $\mathbb{E}X_z = 0$, and Lemma 5 gives subgaussian increments for the metric

$$d(x, y) = C \frac{K}{\sqrt{n}} |x - y|.$$

The standard covering estimate

$$N(S_{\mathbb{C}^k}, |\cdot|, \eta) \leq (3/\eta)^{2k}$$

and Dudley's entropy inequality [3] therefore give

$$\mathbb{E} \sup_{z \in S_{\mathbb{C}^k}} |X_z| \leq CK \sqrt{\frac{k}{n}}. \quad (4)$$

The one-point term is absorbed here because, for fixed z_0 , Lévy concentration gives $\mathbb{E}|X_{z_0}| \leq CL/\sqrt{n} \leq CK\sqrt{k/n}$. Choose the absolute constant in (2) small enough that the right-hand side of (4) is at most $\varepsilon/2$.

Finally,

$$\Phi(U) = \sup_{z \in S_{\mathbb{C}^k}} |f(Uz) - \mu|$$

is L -Lipschitz on the unitary group for its Hilbert–Schmidt geodesic metric. Although the increment calculation used Haar measure on $U(n)$, the random subspace has the same law when generated by Haar measure on $SU(n)$. Concentration on $SU(n)$, in the standard form used in the sharp proof, yields

$$\mathbb{P}\{\Phi > \mathbb{E}\Phi + t\} \leq C \exp(-cnt^2/L^2).$$

Take $t = \varepsilon/2$ and weaken L to K . □

6 $C^{1,1}$ non-circled functions

Corollary 6. *Let F be real-valued and $C^{1,1}$ on a neighborhood of the closed Euclidean unit ball in $\mathbb{C}^n$, considered as a subset of $\mathbb{R}^{2n}$. Suppose*

$$\sup |\nabla F| \leq L, \quad \text{Lip}(\nabla F) \leq H,$$

and put $f = F|_{S_{\mathbb{C}^n}}$. The conclusion of Theorem 1 holds whenever

$$k \leq cn \frac{\varepsilon^2}{(2L + H)^2}.$$

Proof. Let $R_\omega z = \omega z$, viewed as an orthogonal map on $\mathbb{R}^{2n}$. For

$$G_\omega(z) = F(R_\omega z) - F(z)$$

the real-gradient chain rule gives

$$\nabla G_\omega(z) = R_\omega^* \nabla F(R_\omega z) - \nabla F(z).$$

Consequently,

$$\begin{aligned} |\nabla G_\omega(z)| &\leq |\nabla F(R_\omega z) - \nabla F(z)| + |(R_\omega^* - I)\nabla F(z)| \\ &\leq (H + L)|\omega - 1|. \end{aligned}$$

Thus $D_{\mathbb{T}}(f) \leq H + L$, while f is L -Lipschitz. Apply the theorem. $\square$

Remark 7. The corollary includes genuinely non-circled examples. For instance, $F(z) = \operatorname{Re}\langle a, z \rangle$ has $H = 0$ and $D_{\mathbb{T}}(f) = L = |a|$.

7 Limitations and novelty audit

For an arbitrary Lipschitz function, $D_{\mathbb{T}}(f)$ may be infinite: a tiny phase rotation can move a sharp local feature and change the increment's Lipschitz slope at order one. The theorem therefore does not answer the unrestricted source question. It isolates exactly where circledness enters the sharp chaining proof and replaces invariance by a quantitative, checkable regularity condition.

On 23 July 2026, bounded searches covered the exact source question and combinations of *circledness*, *complex Dvoretzky*, *phase increment*, $C^{1,1}$, and *Lipschitz gradient*. The later monograph [2] retains a logarithmic small- ε threshold for arbitrary Lipschitz functions and states that dropping it is unknown. No statement of Theorem 1 or Corollary 6 was located. This is a bounded negative search, so novelty confidence is moderate rather than definitive.

Human-review recommendation

Send to a reviewer familiar with concentration and asymptotic geometric analysis. The key checks are Lemma 4, the conditional mean-zero step in Lemma 3, and the phase-gradient calculation in Corollary 6. If accepted, the result is suitable as a concise proposition complementing the standard circled theorem.

References

- [1] G. Aubrun, S. Szarek, and E. Werner, *Hastings's additivity counterexample via Dvoretzky's theorem*, Comm. Math. Phys. **305** (2011), 85–97; arXiv:1003.4925.
- [2] G. Aubrun and S. Szarek, *Alice and Bob Meet Banach: The Interface of Asymptotic Geometric Analysis and Quantum Information Theory*, Mathematical Surveys and Monographs 223, American Mathematical Society, 2017.
- [3] M. Talagrand, *The Generic Chaining*, Springer Monographs in Mathematics, Springer, 2005.